

Native DS-TWR speed-limit revalidation

Fresh throughput, precision, and raw-range integrity study

Test date: July 31, 2026, EEST
Platform: 5 ESP32-S3 + DW3000 modules, Raspberry Pi 5
Branch: `experiment/passive-ds-twr-pipeline`
Dashboard source: `8244a3432f68`
Active firmware identity: `54a586c-dirty`
Configuration: M1 tag; M2, M3, M4, M5 anchors
Radio: UWB channel 5
Environment: unchanged static indoor setup

Key result. The **60 ms** profile is the fastest setting that completed the decisive 180 s validation with **zero impossible raw ranges**. It delivered **11.63 coherent independent frames/s** and **61.96 rolling display updates/s**, with **1.38 cm precision RMS** and **2.39 cm CEP95**. Relative to the fresh 410 ms control, this is a **6.75× coherent-frame speedup** for only **+1.6% RMS** and **+3.9% CEP95**. The otherwise excellent-looking 59 ms profile was rejected because it produced a physically impossible **−5.301 m raw range**.

Reproducible updated field report

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1 Executive summary

This report revalidates the native three-message DS-TWR speed limit after restoring the low-noise ranging behavior and capturing a new slow control on the same physical setup. The exchange remains strictly POLL-RESP-FINAL. No temporal filter, moving average, position smoothing, or additional radio message is used in the offline comparison.

The study contains 17 capture blocks, 1,860.8 s of data, and 156,935 raw tag-to-anchor ranges. It first repeats the dashboard profiles from 410 ms down to 13 ms, then narrows the transition using 49, 57, 58, 59, 60, 61, and 64 ms candidates. Decisive profiles use 180 s blocks; short 60 s screens are used only to reject obviously unsafe settings or select the next candidate.

Table 1: Main fresh results. Precision is measured around each unfiltered coherent-position series' own mean.

Profile	Coherent	Rolling	Delivery	RMS	CEP95	Raw anomalies
410 ms control	1.72 Hz	9.16 Hz	94.11%	1.36 cm	2.30 cm	0
64 ms reserve	10.92 Hz	58.14 Hz	94.03%	1.38 cm	2.41 cm	0
60 ms recommended	11.63 Hz	61.96 Hz	93.97%	1.38 cm	2.39 cm	0
59 ms rejected	11.82 Hz	62.78 Hz	93.75%	1.31 cm	2.20 cm	1
58 ms rejected	12.00 Hz	63.48 Hz	93.32%	1.35 cm	2.30 cm	2

Decision. Use 60 ms as the fastest currently validated precision profile. Use 64 ms when an additional millisecond of reserve per anchor slot is preferred. Retain 410 ms only as a slow control. Treat every profile below 60 ms as an experimental stress test: their central position distributions can remain tight while rare timestamp failures generate impossible raw distances.

2 Why the recommendation changed

The July 26 report recommended 17 ms and treated 13 ms as the measured maximum. That report faithfully described the complete-position and timing statistics available in its original capture. The present result does not erase that historical data; it changes the operating recommendation for the current firmware and validation criterion.

Three differences matter:

1. the low-noise native DS-TWR implementation was recovered and a fresh 410 ms control was captured on the unchanged current setup;
2. the current dashboard maintains live anchor geometry and distinguishes rolling estimates from strict coherent four-anchor frames;
3. most importantly, every raw range is now audited before solver or frame-assembly rejection. Negative ranges and deviations greater than 10 cm from the per-anchor median are counted even when they do not visibly disturb the position trail.

The new audit found impossible raw measurements at 33, 29, 25, 21, 17, and 13 ms. Their coherent RMS and CEP95 still look comparable to the control. Position precision alone is therefore insufficient to qualify the radio timing. The updated recommendation requires both tight central precision and clean raw timestamp integrity.

3 System and protocol

3.1 Native three-frame DS-TWR

For each anchor, the tag sends POLL, the anchor returns RESP, and the tag sends FINAL. The anchor combines its hardware receive/transmit timestamps with the tag timestamps embedded in the final message and estimates propagation time as

$$\hat{T}_{\text{prop}} = \frac{R_a R_b - D_a D_b}{R_a + D_a + R_b + D_b}, \quad (1)$$

where R_a, R_b are round-trip intervals and D_a, D_b are reply delays. Antenna-delay calibration and the instantaneous carrier-integrator correction remain enabled. They correct physical timestamp behavior and are not temporal position filters.

3.2 Frame and geometry scheduler

The tag visits anchors A2, A3, A4, and A5 sequentially. The base tag-ranging frame is

$$T_{\text{base}} = 4T_{\text{slot}} + T_{\text{gap}}. \quad (2)$$

The current implementation also inserts one rotating anchor-to-anchor geometry-maintenance slot every four base frames. With a 1 ms command delay, the average scheduler period is approximately

$$T_{\text{avg}} = T_{\text{base}} + \frac{T_{\text{slot}} + 1 \text{ ms}}{4}. \quad (3)$$

This maintenance exchange keeps the displayed physical geometry live and is included in measured throughput. It is one reason the measured strict coherent-frame rate is lower than $1/T_{\text{base}}$.

3.3 Coherent frames and rolling updates

A coherent independent position uses four ranges assigned to one complete anchor round. A rolling position is recomputed whenever a new range arrives, using the latest still-fresh ranges from the other anchors. Rolling updates provide low display latency, but adjacent estimates share measurements and must not be interpreted as statistically independent frames.

At 60 ms, the dashboard can produce 61.96 rolling updates/s while the strict offline assembler identifies 11.63 complete independent frames/s. Both rates are reported so display responsiveness is not confused with independent measurement throughput.

4 Experimental method

4.1 Geometry and unchanged setup

All captures use the same cached measured geometry:

Anchor	x [m]	y [m]
A2	0.000	0.000
A3	0.000	3.067
A4	2.971	−0.071
A5	3.196	2.974

The stored tag reference is the anchor centroid, (1.54175, 1.49250) m. Because this coordinate is derived from cached geometry rather than an independent survey of the tag location, the report uses dispersion around each capture mean as its primary precision metric. Absolute bias fields are retained in `summary.json` but are not presented as independently verified absolute accuracy.

Active antenna delays remained M1=16363, M2=16368, M3=16362, M4=16371, and M5=16371. No module was physically moved during the sweep. Runtime profile changes were applied to all five modules in parallel, followed by reboot and warm-up.

4.2 Fresh control and boundary search

The sequence was:

1. capture a new 180 s 410 ms precision control;
2. validate the existing 64, 33, and 29 ms profiles for 180 s and screen 25, 21, 17, and the true 13 ms profile for 60 s;
3. test intermediate 49, 57, 58, 60, and 61 ms candidates for 60 s;
4. perform decisive 180 s validations at 58, 59, and 60 ms.

The short 58 ms screen was clean, but its 180 s validation produced two impossible ranges. Likewise, the 29 ms screen was clean and its long validation failed. These two cases demonstrate why short visual screening cannot establish reliability.

4.3 Acceptance criteria

A candidate is accepted only if:

- the decisive 180 s block contains no negative raw distance;
- no raw distance differs by more than 10 cm from that anchor’s capture median;
- coherent RMS and CEP95 remain within 20% of the 410 ms control;
- all five modules remain online with stable sequence delivery.

The 10 cm threshold is an integrity alarm, not the claimed localization accuracy. Every rejected boundary event was several metres beyond that limit. No range is made acceptable merely because a later solver ignores it.

4.4 Metrics

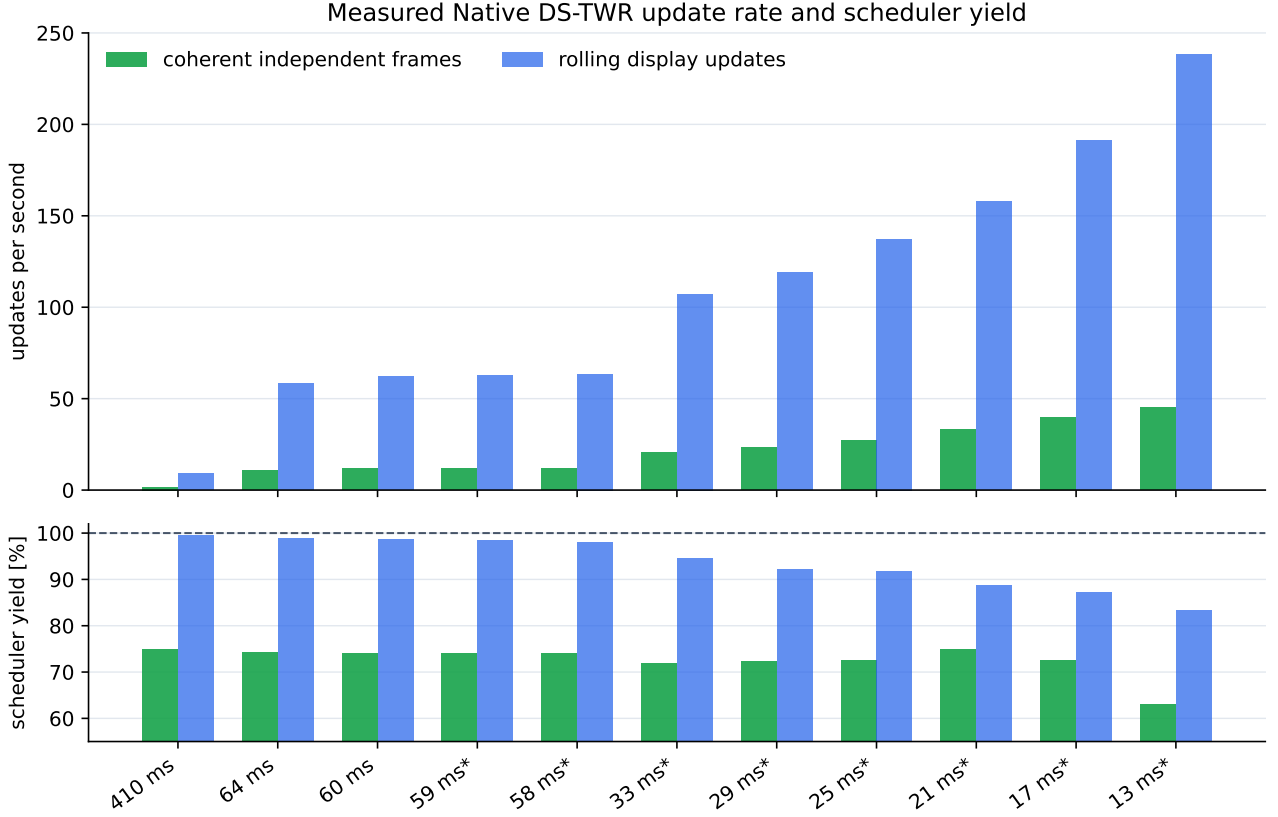
The offline reconstruction uses unweighted all-anchor linear trilateration and no temporal filtering. Precision RMS is

$$\text{RMS}_{\text{precision}} = \sqrt{\frac{1}{N} \sum_{i=1}^N \|\mathbf{p}_i - \bar{\mathbf{p}}\|^2}. \quad (4)$$

CEP50 and CEP95 are the corresponding 50th and 95th percentiles of radial deviation from the capture mean. Sequence delivery is the fraction of observed 16-bit range sequences within the capture span. Raw integrity is calculated directly from every `UWB_RANGING result` record.

5 Results

5.1 Update rate and scheduler yield



* Raw-integrity validation failed: at least one impossible distance was observed.

Figure 1: Measured coherent and rolling update rates, plus the yield of each stream relative to its ideal average scheduler rate. An asterisk marks a profile that failed raw-integrity validation. Colors retain the metric meanings shown in the legend.

The 60 ms profile yields a $6.75\times$ coherent-frame speedup and a $6.77\times$ rolling-update speedup over the 410 ms control. Sequence delivery changes by only -0.14 percentage points. The lower timing profiles raise apparent update rate, but raw integrity fails. In the lower panel, coherent yield is the observed independent-frame rate divided by the ideal geometry-adjusted frame rate. Rolling yield uses four ideal display updates per complete tag-ranging frame. The difference quantifies how shared fresh-range updates remain responsive even when strict four-range frames are incomplete.

5.2 Central precision

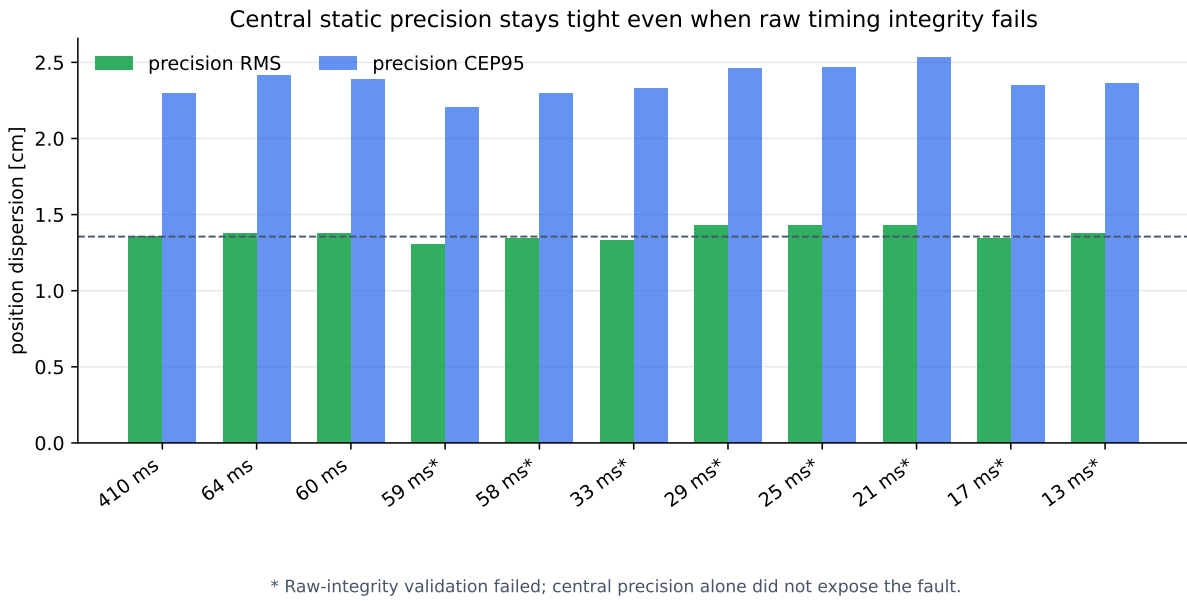


Figure 2: Unfiltered coherent-position dispersion. The rejected profiles are marked with an asterisk but still have deceptively good RMS and CEP95. Colors retain the RMS/CEP95 meanings shown in the legend.

The 60 ms coherent RMS is 1.3766 cm, compared with 1.3555 cm at 410 ms: +0.0211 cm or +1.56%. CEP95 changes from 2.2987 to 2.3892 cm: +0.0905 cm or +3.94%. These are small differences relative to ordinary between-capture variation.

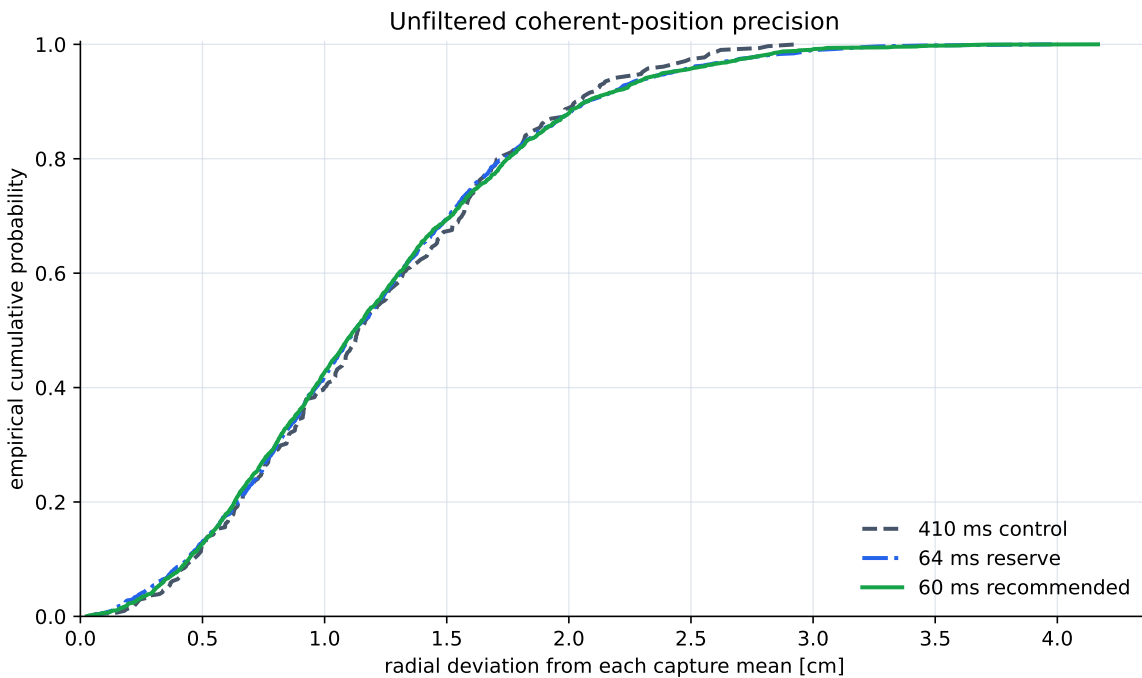


Figure 3: Empirical CDF of radial coherent-position deviation from each capture mean. The accepted 410, 64, and 60 ms distributions nearly overlap.

The CDF confirms that the recommended 60 ms setting preserves the complete shape of the static precision distribution rather than only its RMS summary.

5.3 Raw-range integrity

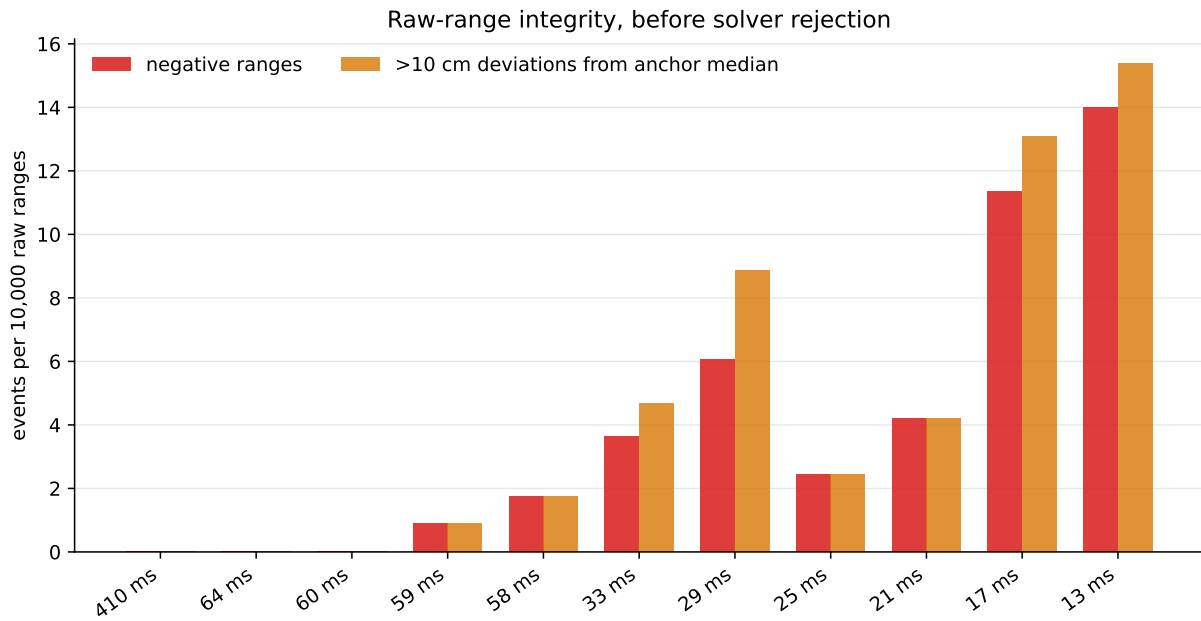


Figure 4: Impossible measurements per 10,000 raw ranges, counted before any frame or solver rejection.

The accepted 410, 64, and 60 ms captures contain no raw anomaly. Every existing dashboard profile below 60 ms produces at least one impossible measurement in its fresh capture. At 17 ms, 13 negative ranges and 15 total spikes appear in 11,467 observations; at 13 ms, the counts rise to 20 and 22 in 14,303 observations.

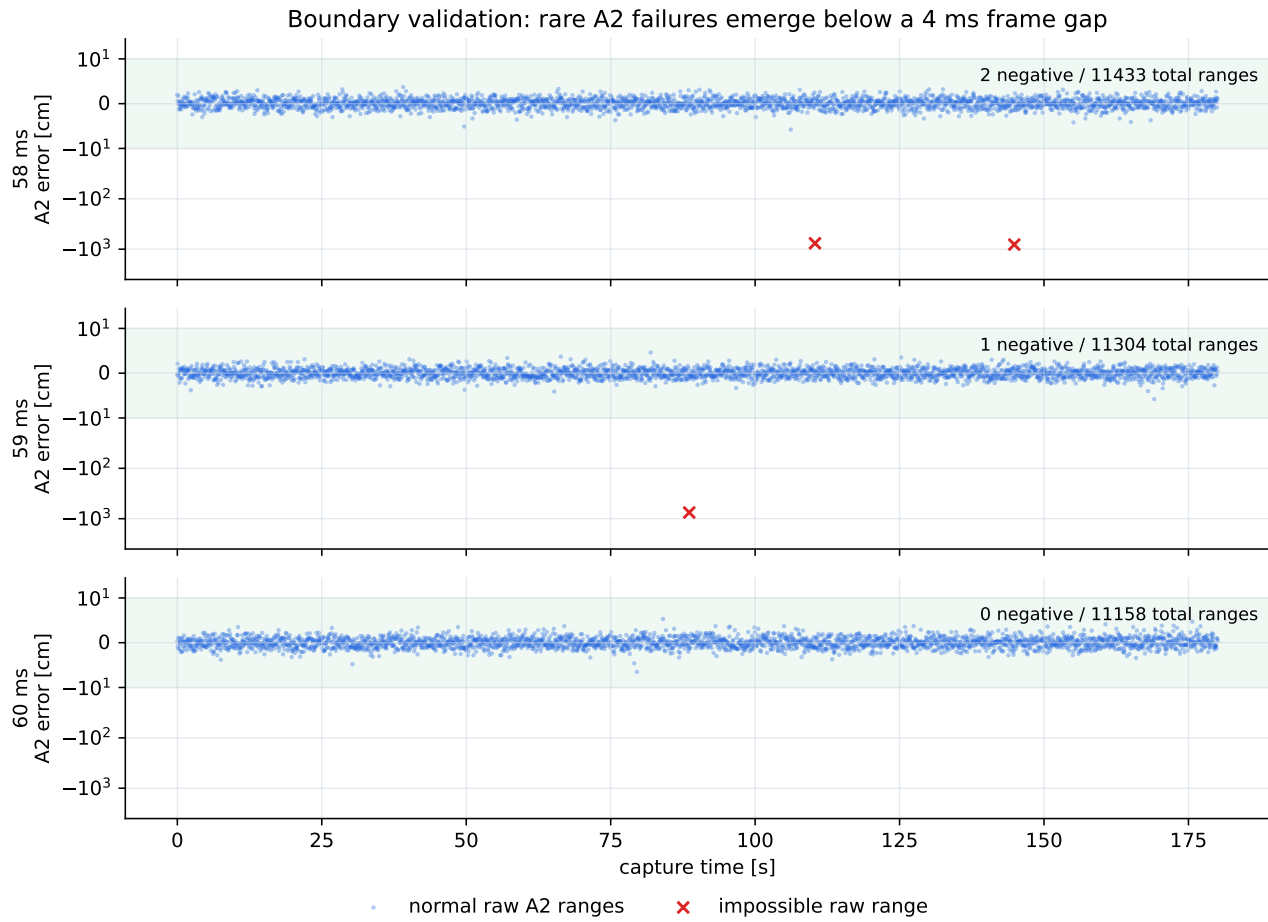


Figure 5: All A2 raw deviations during the decisive 58, 59, and 60 ms validations. The symlog axis preserves centimetre-scale noise and metre-scale failures in the same figure.

The boundary result is unusually sharp. The 58 ms profile produced two negative A2 ranges, the 59 ms profile produced one, and the otherwise identical 60 ms profile produced none. All three use 14 ms anchor slots, 8 ms RX timeout, and 2+2 ms response/final delays. Only the frame gap changes: 2, 3, and 4 ms respectively.

5.4 Accepted per-anchor stability

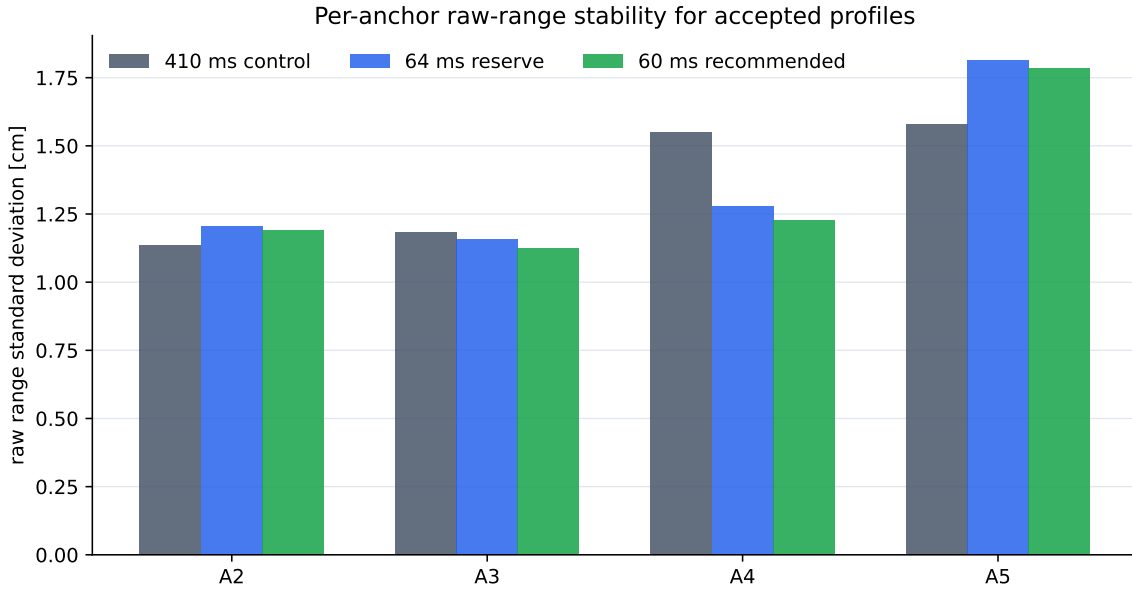


Figure 6: Raw range standard deviation for the three accepted profiles.

The accepted profiles retain approximately 1–2 cm raw standard deviation per anchor. The 60 ms values are 1.19 cm (A2), 1.12 cm (A3), 1.23 cm (A4), and 1.79 cm (A5). There is no broad noise inflation as the frame period falls from 410 to 60 ms.

6 Interpretation and recommendation

6.1 Observed limiting mechanism

The data do not show a gradual accuracy-versus-speed tradeoff near the boundary. Central RMS is actually slightly lower at 59 ms than at 60 ms. Instead, rare metre-scale failures appear while the ordinary range population remains centimetre-stable.

Because 58, 59, and 60 ms use the same 14 ms slot and exchange delays, while only the inter-frame gap changes, the evidence strongly implicates insufficient recovery margin between rounds. Every decisive boundary failure is emitted by A2, the first anchor in the configured list. This pattern is consistent with a round-boundary timestamp or radio-state race, but the present field data do not prove the exact firmware instruction or DW3000 state responsible.

The correct engineering response is therefore to retain a clean timing margin, not to add a position filter that hides the symptom. Instrumenting the round transition and validating an explicit timestamp sanity check remain useful future firmware tasks.

6.2 Recommended profiles

Default: 60 ms precision-speed limit. Use 14 ms slots, 4 ms frame gap, 8 ms RX timeout, 2 ms response delay, 2 ms final delay, 100 ms anchor RX slice, and 500 UUS auto-RX delay. This is the fastest profile with a clean 180 s raw audit.

Reserve: 64 ms. Use 15 ms slots and the same 4/8/2/2 ms gap/timeout/response/final values. It loses about 6% coherent throughput relative to 60 ms but also passed its 180 s capture with zero raw anomalies.

Control: 410 ms. Use 100 ms slots, 10 ms gap, 90 ms timeout, and 20+20 ms delayed transmissions only when reproducing the slow precision baseline. It is no longer justified as the normal operating profile.

Experimental profiles below 60 ms. They remain available in the dashboard for controlled investigation, but are now marked rejected or experimental. Their high rolling-update numbers must not be presented as validated reliable positioning throughput.

7 Limits of inference and next tests

This is a strict static revalidation at one favorable point, not a complete localization qualification. The 60 ms decision applies to the tested modules, firmware state, channel, anchor order, geometry, and host load. It does not yet establish:

- multi-hour probability of a rare boundary failure;
- performance while the tag translates or rotates;
- behavior near or outside the anchor perimeter;
- NLOS, human blockage, or severe multipath performance;
- sensitivity to Wi-Fi/dashboard load, temperature, or supply voltage;
- behavior with a different first anchor or additional modules.

The next validation should run 60 and 64 ms for at least one hour, alternate their order, and repeat with A2 moved away from the first slot. A controlled dynamic trajectory should then compare coherent latency and precision. If sub-60 ms operation remains important, add per-stage round-transition timestamps and an explicit hardware-timestamp validity guard before changing the estimator or adding filters.

A Complete profile table

Table 2: Fresh aggregate results. Durations are 180 s unless marked 60 s screen.

Frame	Duration	Coherent	Rolling	Delivery	RMS	CEP95	Neg.	Spikes
410	180 s	1.72	9.16	94.11%	1.36	2.30	0	0
64	180 s	10.92	58.14	94.03%	1.38	2.41	0	0
60	180 s	11.63	61.96	93.97%	1.38	2.39	0	0
59	180 s	11.82	62.78	93.75%	1.31	2.20	1	1
58	180 s	12.00	63.48	93.32%	1.35	2.30	2	2
57	60 s	12.21	64.37	93.02%	1.38	2.42	1	1
49	60 s	17.61	74.26	92.46%	1.30	2.25	1	1
33	180 s	20.42	107.21	90.61%	1.33	2.33	7	9
29	180 s	23.37	119.09	88.92%	1.43	2.46	13	19
25	60 s	27.09	137.28	89.28%	1.43	2.47	2	2
21	60 s	33.34	157.85	87.90%	1.43	2.53	4	4
17	60 s	39.77	191.04	87.14%	1.34	2.35	13	15
13	60 s	44.99	238.12	85.30%	1.38	2.36	20	22

Coherent and rolling columns are updates/s. RMS and CEP95 are centimetres. Spikes count all raw distances more than 10 cm from the corresponding per-anchor median and include the negative observations.

B Timing profiles

Frame	Slot	Gap	RX timeout	RESP delay	FINAL delay
410	100	10	90	20	20
64	15	4	8	2	2
60	14	4	8	2	2
59	14	3	8	2	2
58	14	2	8	2	2
57	14	1	8	2	2
49	12	1	8	2	2
33	8	1	5	2	2
29	7	1	5	2	2
25	6	1	5	2	2
21	5	1	5	2	2
17	4	1	4	1	1
13	3	1	3	1	1

Table 3: Runtime values in milliseconds. RX slice is 100 ms and automatic RX delay is 500 UUS for every row.

C Files and reproduction

File/directory	Contents
README.md	fresh 410 ms control identity and precision reference
PROFILE_REVALIDATION.md	concise findings and boundary decision
profile_comparison.csv	machine-readable aggregate table
/data/.jsonl	raw range/status events and timing logs
/data/.metadata.json	full initial/final module status and runtime configuration
*/positions.csv	unfiltered coherent and rolling reconstructions
*/summary.json	complete per-capture metrics
*/raw_quality.json	per-anchor raw integrity audit
figures/	report plots in vector PDF and PNG

Reproduction

```
cd /home/pi/Documents/UWB
python3 tools/uwb_native_ds_speed_revalidation_report.py
cd reports/native_ds_speed_revalidation_20260731
latexmk -pdf -interaction=nonstopmode -halt-on-error report.tex
```

References

- [1] Qorvo, *DW3xxx Device Driver API Guide*, double-sided two-way-ranging examples, 2025. <https://forum.qorvo.com/uploads/short-url/xD3TlXKvkujjdUaJWxv2b4E7GQN.pdf>.
- [2] Decawave, *APS013: DW1000 and Two-Way Ranging*, rev. 2.3, Poll-Response-Final and the asymmetric DS-TWR formula. <https://forum.qorvo.com/uploads/short-url/x34DrF7EW5fQP9wY3aNESqPKz8z.pdf>.
- [3] Local repository *UWB Module ESP-IDF*, branch `experiment/passive-ds-twr-pipeline`, dashboard source `8244a3432f68caf23d604e475cbe718698c4ab2f`, active module firmware identity `54a586c-dirty`, field state from July 31, 2026.